

DATABASE PROJECT REPORT

Project #4: University Research & Lab Asset Tracker

SafeLab: An Integrated System for Research Grant, Equipment,
Chemical Inventory, and Access Control Management

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Database Design & Requirements Report

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1. Introduction & Project Scope

Standards basis: Adapted from ISO/IEC/IEEE 29148 for requirements and system scope documentation.

1.1 System Objective

The University Research & Lab Asset Tracker (hereafter SafeLab) is designed to give a university research division a single system of record for everything that moves through its laboratories: the grants that fund the work, the equipment purchased under those grants, the chemical inventory stored on-site, and the people — students, faculty, and staff — who interact with all of the above.

The system's primary objectives are:

1. **Financial traceability** — Link every piece of equipment and every chemical purchase back to the research grant that funded it, so that grant expenditure can be audited against the funding agency's terms.
2. **Asset visibility** — Provide real-time status (available, in use, under maintenance, retired) for every tracked piece of equipment across all labs in a department.
3. **Safety compliance** — Guarantee, at the database level, that hazardous chemicals and controlled equipment can only be accessed, checked out, or logged as entered by personnel holding a current, unexpired safety certification appropriate to that hazard class. This core rule is enforced with SQL triggers rather than relying on application-layer checks alone.
4. **Accountability** — Maintain an immutable audit trail (access log) of who accessed which lab, equipment, or chemical, and when, including access attempts that were denied for lack of certification.
5. **Lifecycle management** — Track certification issue/expiry dates, chemical expiry dates, and equipment maintenance/retirement dates so the system can proactively flag anything that is about to lapse.

1.2 Business Rules & Constraints

Rule	Business Rule / Constraint
BR-1	Every piece of equipment must be associated with exactly one lab and, if purchased with grant funding, with exactly one research grant.
BR-2	A research grant must have a designated Principal Investigator (PI), who must be a person with role faculty.
BR-3	A chemical inventory item with a hazard class other than none must specify a minimum required certification.
BR-4	Hazardous materials may only be accessed (checked out, viewed, or physically entered into storage) by personnel holding a valid, non-expired certification matching the chemical's required hazard class. Any access attempt by uncertified personnel must be rejected at the database level and still recorded in the access log as <code>authorized = FALSE</code> .
BR-5	A certification expires automatically after its defined validity period; the system must never treat an expired certification as valid, even if the row still exists in the certification-history table.
BR-6	Equipment reservations must not overlap for the same piece of equipment (no double-booking).
BR-7	A grant's cumulative equipment/chemical purchase value must not exceed its approved budget.
BR-8	A chemical inventory item past its expiry date may not be checked out for use (only disposal-related access is permitted).
BR-9	Only users with the <code>lab_manager</code> or <code>admin</code> database role may insert, update, or delete rows in <code>Certification</code> , <code>PersonnelCertification</code> , or <code>ChemicalInventory</code> .
BR-10	The access log is append-only: no <code>UPDATE</code> or <code>DELETE</code> is permitted on <code>AccessLog</code> by any application role, to preserve

2. Database Design

Standards basis: Conceptual and logical database design is documented with reference to ISO/IEC 19505 and relevant relational design practices.

2.1 Conceptual Model (ER/EER Diagram)

The SafeLab database contains the following core entities:

- Department
- Personnel (faculty, postdoc, grad_student, undergrad_student, staff)
- Lab
- ResearchGrant
- Certification
- PersonnelCertification (associative entity: many-to-many between Personnel and Certification, time-stamped)
- Equipment
- ChemicalInventory
- EquipmentReservation
- AccessLog

Key relationships:

- Department 1 — N Lab
- Department 1 — N Personnel
- ResearchGrant N — 1 Personnel (PI, role = faculty)
- Lab 1 — N Equipment
- Lab 1 — N ChemicalInventory
- ResearchGrant 1 — N Equipment (equipment purchased under a grant, optional)
- ResearchGrant 1 — N ChemicalInventory (optional)
- Personnel M — N Certification (via PersonnelCertification)
- Certification 1 — N ChemicalInventory (min_required_cert_id — certification needed to legally access that chemical)
- Personnel 1 — N EquipmentReservation; Equipment 1 — N EquipmentReservation
- Personnel 1 — N AccessLog; Lab 1 — N AccessLog

ER Diagram — Mermaid Source

erDiagram

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DEPARTMENT ||--o{ LAB : contains
DEPARTMENT ||--o{ PERSONNEL : employs
PERSONNEL ||--o{ RESEARCH_GRANT : "is PI of"
LAB ||--o{ EQUIPMENT : houses
LAB ||--o{ CHEMICAL_INVENTORY : stores
RESEARCH_GRANT ||--o{ EQUIPMENT : funds
RESEARCH_GRANT ||--o{ CHEMICAL_INVENTORY : funds
CERTIFICATION ||--o{ CHEMICAL_INVENTORY : "required for"
PERSONNEL ||--o{ PERSONNEL_CERTIFICATION : holds
CERTIFICATION ||--o{ PERSONNEL_CERTIFICATION : "granted as"
PERSONNEL ||--o{ EQUIPMENT_RESERVATION : books

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EQUIPMENT ||--o{ EQUIPMENT_RESERVATION : "is booked in"
 PERSONNEL ||--o{ ACCESS_LOG : generates
 LAB ||--o{ ACCESS_LOG : records

Cardinality notes: The ResearchGrant → Equipment/ChemicalInventory link is optional (0..N) because a lab may also hold department-owned assets not tied to any grant. The Certification → ChemicalInventory link is mandatory only when hazard_class <> 'none'.

2.2 Logical Schema Mapping

Entity / Relationship	Relational Table	Primary Key	Foreign Keys
Department	department	dept_id	—
Personnel	personnel	person_id	dept_id → department
Lab	lab	lab_id	dept_id → department
ResearchGrant	research_grant	grant_id	pi person_id → personnel
Personnel–Certification (M:N)	personnel_certification	(person_id, cert_id, issue_date)	person_id → personnel; cert_id → certification
Equipment	equipment	equipment_id	lab_id → lab; grant_id → research_grant
ChemicalInventory	chemical_inventory	chemical_id	lab_id → lab; grant_id → research_grant; min_required_cert_id → certification
EquipmentReservation	equipment_reservation	reservation_id	equipment_id → equipment; person_id → personnel
AccessLog	access_log	log_id	person_id → personnel; lab_id → lab

2.3 Normalization Verification (1NF → 3NF/BCNF)

Starting Point — Unnormalized Flat Record

A typical spreadsheet-style equipment record might contain duplicated lab, department, grant, PI, and certification information:

Equipment(equipment_id, equipment_name, lab_id, lab_name, dept_id, dept_name, grant_id, grant_title, pi_name, cert_names_required)

Step 1 — First Normal Form (1NF)

cert_names_required is a repeating/multi-valued group. It is removed into its own row-per-value table, so every attribute has an atomic value:

Equipment(equipment_id, equipment_name, lab_id, lab_name, dept_id, dept_name, grant_id, grant_title, pi_name)

EquipmentCertRequirement(equipment_id, cert_id)

Result: the relations satisfy 1NF.

Step 2 — Second Normal Form (2NF)

Equipment has a single-column primary key (equipment_id), so there can be no partial dependency on part of a composite key. EquipmentCertRequirement(equipment_id, cert_id) is fully dependent on its whole composite key. Therefore, both relations satisfy 2NF.

Step 3 — Third Normal Form (3NF)

Functional dependencies in the original Equipment relation include:

- equipment_id → lab_id, lab_name, dept_id, dept_name, grant_id, grant_title, pi_name
- lab_id → lab_name, dept_id, dept_name
- grant_id → grant_title, pi_name

The second and third dependencies create transitive dependencies because lab and grant attributes depend on non-key attributes rather than directly on equipment_id. The schema is therefore decomposed as follows:

Equipment(equipment_id, equipment_name, lab_id, grant_id)

Lab(lab_id, lab_name, dept_id)

Department(dept_id, dept_name)

ResearchGrant(grant_id, grant_title, pi_person_id)

Personnel(person_id, pi_name, ...)

Result: every non-key attribute depends on the key, the whole key, and nothing but the key; 3NF is achieved.

Step 4 — BCNF Check

- Equipment: equipment_id → equipment_name, lab_id, grant_id. equipment_id is the sole candidate key.
- Lab: lab_id → lab_name, dept_id. lab_id is the sole candidate key.
- PersonnelCertification(person_id, cert_id, issue_date) → expiry_date: the determinant is the composite candidate key.
- ChemicalInventory: chemical_id → name, cas_number, hazard_class, lab_id, min_required_cert_id, expiry_date. chemical_id is the sole key, and cas_number is also a candidate key (UNIQUE).

All listed determinants are superkeys in their respective relations. Therefore, the decomposed relations satisfy BCNF, with no remaining insertion, update, or deletion anomalies caused by the identified dependencies.

3. Data Dictionary

Adapted from ISO/IEC 11179 principles for metadata and data-element documentation.

Table: department

Attribute	Type	Constraints	Description
dept_id	INT	PK, AUTO INCREMENT	Unique department identifier
dept_name	VARCHAR(100)	NOT NULL, UNIQUE	Department name
building	VARCHAR(50)	NOT NULL	Building housing the department

Table: personnel

Attribute	Type	Constraints	Description
person_id	INT	PK, AUTO INCREMENT	Unique person identifier
first_name	VARCHAR(50)	NOT NULL	First name
last_name	VARCHAR(50)	NOT NULL	Last name
email	VARCHAR(120)	NOT NULL, UNIQUE	University email
role	ENUM('faculty','postdoc','grad_student','undergrad_student','staff')	NOT NULL	Role determines default access privileges
dept_id	INT	FK → department	Home

			department
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Table: research_grant

Attribute	Type	Constraints	Description
grant_id	INT	PK, AUTO INCREMENT	Unique grant identifier
title	VARCHAR(200)	NOT NULL	Grant title
funding_agency	VARCHAR(150)	NOT NULL	Sponsoring agency
budget	DECIMAL(12,2)	NOT NULL, CHECK (budget > 0)	Approved budget (USD)
start_date	DATE	NOT NULL	Grant start date
end_date	DATE	NOT NULL, CHECK (end_date > start_date)	Grant end date
pi_person_id	INT	FK → personnel, NOT NULL	Principal Investigator

Table: lab

Attribute	Type	Constraints	Description
lab_id	INT	PK, AUTO INCREMENT	Unique lab identifier
lab_name	VARCHAR(100)	NOT NULL	Lab name
dept_id	INT	FK → department, NOT NULL	Owning department
room_number	VARCHAR(20)	NOT NULL	Physical room
biosafety_level	SMALLINT	CHECK (biosafety_level BETWEEN 1 AND 4)	BSL rating of the lab

Table: certification

Attribute	Type	Constraints	Description
cert_id	INT	PK, AUTO_INCREMENT	Unique certification identifier
cert_name	VARCHAR(100)	NOT NULL	e.g. Radioactive Materials Handling
hazard_class	VARCHAR(50)	NOT NULL	Hazard category covered by the certification
valid_period_months	SMALLINT	NOT NULL, CHECK (valid_period_months > 0)	Validity length

Table: personnel_certification

Attribute	Type	Constraints	Description
person_id	INT	PK(part), FK → personnel	Certified person
cert_id	INT	PK(part), FK → certification	Certification held
issue_date	DATE	PK(part), NOT NULL	Date issued
expiry_date	DATE	NOT NULL	Computed/stored expiry (issue_date + valid_period_months)

Table: equipment

Attribute	Type	Constraints	Description
equipment_id	INT	PK, AUTO INCREMENT	Unique equipment identifier
equipment_name	VARCHAR(150)	NOT NULL	Name/model
category	VARCHAR(80)	NOT NULL	e.g. Centrifuge, Spectrometer

lab_id	INT	FK → lab, NOT NULL	Home lab
grant_id	INT	FK → research_grant, NULLABLE	Funding grant, if any
purchase_date	DATE	NOT NULL	Purchase date
value	DECIMAL(10,2)	NOT NULL, CHECK (value >= 0)	Purchase value
status	ENUM('available','in_use','maintenance','retired')	NOT NULL DEFAULT 'available'	Current status

Table: chemical_inventory

Attribute	Type	Constraints	Description
chemical_id	INT	PK, AUTO_INCREMENT	Unique chemical record identifier
name	VARCHAR(150)	NOT NULL	Chemical name
cas_number	VARCHAR(20)	UNIQUE	CAS registry number
hazard_class	VARCHAR(50)	NOT NULL DEFAULT 'none'	e.g. flammable, carcinogen, radioactive, none
quantity	DECIMAL(10,3)	NOT NULL, CHECK (quantity >= 0)	Amount on hand
unit	VARCHAR(20)	NOT NULL	e.g. L, kg
lab_id	INT	FK → lab, NOT NULL	Storage lab
grant_id	INT	FK → research_grant, NULLABLE	Funding grant, if any
storage_location	VARCHAR(100)	NOT NULL	Cabinet/shelf code
expiry_date	DATE	NOT NULL	Expiry date
min_required_cert_id	INT	FK → certification, NULLABLE	Certification required to access; NULL if hazard_class = none

Table: equipment_reservation

Attribute	Type	Constraints	Description
reservation_id	INT	PK, AUTO_INCREMENT	Unique reservation identifier
equipment_id	INT	FK → equipment, NOT NULL	Reserved equipment
person_id	INT	FK → personnel, NOT NULL	Reserving person
start_time	TIMESTAMP	NOT NULL	Reservation start
end_time	TIMESTAMP	NOT NULL, CHECK (end_time > start_time)	Reservation end
purpose	VARCHAR(200)	NULLABLE	Stated purpose

Table: access_log

Attribute	Type	Constraints	Description
log_id	BIGINT	PK, AUTO_INCREMENT	Unique log entry
person_id	INT	FK → personnel, NOT NULL	Person attempting access
lab_id	INT	FK → lab, NOT NULL	Lab where access occurred
resource_type	ENUM('equipment','chemical','lab_entry')	NOT NULL	Type of resource accessed
resource_id	INT	NULLABLE	equipment_id or chemical_id, if applicable
access_time	TIMESTAMP	NOT NULL DEFAULT CURRENT_TIMESTAMP	When access occurred/was attempted
access_type	ENUM('checkout','checkin','view','entry')	NOT NULL	Nature of the access

authorized	BOOLEAN	NOT NULL	Whether the system permitted the access
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4. Conclusion

SafeLab provides a centralized relational database design for managing university research grants, laboratory equipment, chemical inventory, personnel certifications, reservations, and access auditing. The design separates independent entities and their relationships to achieve 3NF/BCNF, while the stated business rules establish the foundation for database-level enforcement of financial, safety, authorization, and audit requirements.

In particular, the design emphasizes safety and accountability: certification validity, hazardous-material access, chemical expiry, reservation conflicts, grant-budget limits, role-based data modification, and append-only audit logging should be enforced through appropriate SQL constraints, triggers, privileges, and transactional logic in the implementation phase.